Name: Bosan Hsu

Detroit Tigers Baseball Analytics Questionnaire

Upload your responses to box using this link. Include your full name in all filenames. If you have any issues or questions, please reach out to analyticsstaffing@tigers.com. Please adhere to the word limits.

1. Please give a brief overview of your experience with predictive modeling, as well as a description of an analytical project that you have completed. Include the purpose of the project, the methods or models chosen, any additional methods or models that were tested, and what tools or programming languages you used to complete the project. This project does not need to relate to baseball. [250 words]

My representative predictive modeling project is "Fantasy Baseball Draft - Reinforcement Learning". The objective was to maximize the head-to-head win rate under roster and draft constraints.

I scraped **FanGraphs** and used **Python**/pandas to merge hitters' batting and fielding data. I defined ten-hitter and ten-pitcher categories, standardized features (z-scores), and used **PCA** to form composite strength scores. Analysis showed **positional scarcity** at C and 2B.

I tested draft strategies:

(1) Pick the highest PCA score
(2) PCA score weighted by remaining positional needs
(3) Target positions with the largest strong–weak gap.

I ran 500 simulations and used **OLS regression** (quick read on direction and magnitude) to compare strategies. Strategy 3 performed best.

I then trained a Deep Q-Network (DQN) that observes team state (category totals, available players, round/order) and chooses among S1/S2/S3 each pick; it outperformed static strategies by 7.3% in held-out simulations.

Reflection:

Excessive computation limited my ability to scale an idea: drafting extreme specialists in a single standout category and systematically comparing that to balanced, composite approaches. I would also report a clear per-category z-score baseline alongside PCA for transparency.

Baseball application:

I would segment players into archetypes (e.g., contact-first, power-only, OBP engines) using clustering and attach strength tiers within each archetype. Then, under real constraints (e.g., budget, roster spots, positional scarcity), I'd train an RL policy to choose which archetype/strength to acquire at each step. Finally, I'd run multiple simulations to compare archetype mixes and produce the budget-aware roster policy that maximizes projected win rate.

2. Describe a meaningful experience you have had working or playing in a team setting (does not have to be in sports). [250 words]

Being the captain of my Economics department's basketball team was a meaningful experience. We had no coach, only eight players, and a very young roster, mostly freshmen with little full-court experience. My task was to keep us competitive while developing the younger players.

Actions:

First, as a junior, I discussed the minutes-played plan with the seniors, and we agreed that in scrimmages, we would **give more minutes to freshmen**. I did **1-on-1 check-ins** with each freshman to set personal small goals (like 5 rebounds, 2 steals, or better conditioning) and a plan to reach them. Third, I ran a **weekly film session** for freshmen. We watched full-court clips **frame by frame**. I explained the situation and the options, then asked them to **say the decision back in their own words** so we could spot any gaps. In games, we talked about urgent issues right away. Other feedback waited for the film, so emotions wouldn't get in the way.

Results:

Sadly, we were eliminated, but every player created **open shots** from good spacing and movement, and **everyone scored**. Several freshmen **hit their personal goals**. Teammates said they felt **clearer about what to do** on the court.

I learned:

1. Set clear roles and goals through honest one-on-one talks.
2. Build quick, respectful feedback loops by splitting "in-game now" vs. "film later" and using the say-it-back check.

These habits carry into any teamwork: set clear roles, turn debate into simple metrics, and share plans that a cross-functional group can execute under pressure.

3. Choose a major or minor league baseball player and evaluate their value to their organization, along with the skills that drive their production. Use data and sources to support your reasoning. If selecting a younger player, include both current ability and projected peak. Your goal is to make the evaluation comprehensive, transparent, and grounded in evidence. Avoid players whose value is universally accepted; your analysis should reflect independent thinking. [250 words]

In 2025, Ivan Herrera delivered an xwOBA of .380, a hard-hit rate of 47.7%, and a barrel rate of 11.0%. On FanGraphs, his 2025 line shows a wRC+ of 137 and roughly 2.7 fWAR, indicating that his great contact skills are converting into real production. Most of his workload came at DH, with only 14 games at catcher, which depresses fWAR via the DH positional penalty, making the total more impressive given the bat.

From 2023 to 2025, his batted-ball shape steadily improved: hard-hit and barrel rates rose, ground-ball rate fell, and opposite-field contact ticked up. His plate discipline also improved, with better zone-contact and in-zone swing rates. He also produced positive run values against fastballs, sliders, and sinkers, showing better pitch recognition and consistent contact quality. Taken together, these are stabilizing indicators rather than a one-year blip.

Herrera's defense remains the limiter (pop time/throwing, blocking, running-game control), but the Cardinals publicly said they want him catching again in 2026, with a winter program focused on throwing and game-calling. If he holds this bat while catching regularly, the catcher positional boost (vs. the DH penalty) should raise his WAR baseline. For the organization, an above-average hitter at catcher is scarce and materially raises lineup depth and roster flexibility.

Catchers usually reach their peak around age 27-29, so at 25, Herrera is entering that window. On this trajectory, 3~4 WAR peak seasons are a realistic outcome, with upside if the defense trends toward average.

4. Attached are pitch-level data from two games (AnalyticsQuestionnairePitchData.csv). Please produce self-explanatory graphs and tables to summarize the players' performance in those two games. Summarize your findings with bullet points. Include all code/workbooks.

Pitcher Evaluation

Reliever samples are too small to rate (most 1.0 IP, none over 2.0). I'm skipping prose and showing their metrics in the HTML tables.

For future reports, I'll use the table below to evaluate player performance.

Metric	Average SP	Quality SP	Top SP
IP per start	5.2~5.8	6.0~6.5	≥6.5~7.0
Pitches per BF	3.9~4.2	3.7~3.9	≤3.6
K%	21~24%	25~29%	≥30%
BB%	7~8%	5~6%	≤4~5%
K - BB% (pct points)	12~16	18~22	≥23~25
WHIP	1.25~1.35	1.10~1.20	≤1.05~1.00
OPS allowed	.700~.730	.640~.680	≤.600~.630
First-Pitch Strike%	58~60%	61~64%	≥65%
CSW%	27~29%	29~31%	≥31~32%
Whiff% (per swing)	24~27%	28~31%	≥32%
In-Play% (per swing)	33~44%	26~33%	≤24~26%
Contact% (per swing)	73~76%	70~73%	≤69~70%

Primary Starter: Two Starts (Pitcher 1)

Overview (Pitcher)

Box Score

Pitches	Balls	Strikes	H	HR	BB	K	IP	WHIP	OPS
198	66	101	8	1	3	13	12.0	0.92	0.529

Advanced Stats

Batter Faced	Pitches/BF	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	K%	BB%
47	4.21	48.9%	20.7%	46.5%	15.2%	33.7%	84.8%	27.7%	6.4%

Advanced Stats vs RHH

Batter Faced	Pitches/BF	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	K%	BB%
23	4.26	43.5%	23.5%	46.9%	19.6%	32.6%	80.4%	34.8%	0.0%

Advanced Stats vs LHH

Batter Faced	Pitches/BF	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	K%	BB%
24	4.17	54.2%	18.0%	46.0%	10.9%	34.8%	89.1%	20.8%	12.5%

Note: Per pitch: CSW%, Swing%; Per swing: Whiff%, In-Play%, Contact%; Per PA: K%, BB%, K–BB%

- **Game Stats**
  - **Elite:** 0.92 WHIP, 0.529 OPS,
  - **Quality:** 6 IP/G, 27.7% K%, 6.4% BB%, 21.3% K%-BB%
  - **Average:** 4.21 Pitches/BF, 33.7 In-Play%
  - **Below Average:** 48.9 FPS%, 20.7% CSW, 15.2 Whiff%, 80.4 Contact%

- **R/L Split (directionally meaningful; 2-game sample):**
  - K%: 34.8% vs RHH → 20.8% vs LHH (−14 pts).
  - BB%: 0.0% vs RHH → 12.5% vs LHH (+12.5 pts).
  - Whiff% (per swing): 19.6% → 10.9% (−8.7 pts).
  - CSW% (per pitch): 23.5% → 18.0% (−5.5 pts).
  - Contact% (per swing): 80.4% → 89.1% (+8.7 pts).
  - First-pitch strike%: 43.5% vs RHH (low) vs 54.2% vs LHH (higher)
    - Lefties still drew more walks and fewer Ks → issue is finishing ABs vs LHH, not starting counts.
- **In short:**
  - Results (WHIP & OPS) were excellent; the K-BB% profile is solid.
  - Process signals:
    - Miss rates are light (low CSW/Whiff), and FPS% is below target

        - In-play rate per pitch is only average, so the low WHIP/OPS likely reflects weak contact/sequence/defense in this sample.
    - Future focus: push FPS% toward 55~60%.

Pitch Mix & Results

Pitch Type: All Min sample (Pitches ≥): 0 Update

First row is TOTAL (pitcher-level). Below are per pitch type.

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
TOTAL	198	—	48.9%	20.7%	46.5%	15.2%	33.7%	84.8%	40.9%	31.8%
FF	87	43.9%	42.1%	18.4%	44.8%	5.1%	28.2%	94.9%	43.7%	33.3%
SL	32	16.2%	45.5%	34.4%	50.0%	50.0%	31.2%	50.0%	37.5%	31.2%
CH	30	15.2%	100.0%	20.0%	53.3%	12.5%	56.2%	87.5%	46.7%	23.3%
CU	27	13.6%	44.4%	25.9%	25.9%	14.3%	42.9%	85.7%	37.0%	40.7%
FC	22	11.1%	66.7%	4.5%	63.6%	7.1%	21.4%	92.9%	31.8%	27.3%

- **Pitch Mix & Result**
    - **FF** (44% use): Average strike entry with low Whiff%, not a put-away pitch.
    - **SL** (16%): Pitcher 1's highest Whiff% pitch.
    - **CH** (15%): Lowest Ball% but gets high In-Play%. Fine for strike-getting, risky with traffic.
    - **CU** (14%): The highest Ball%, the lowest Whiff%, and a 42.9 In-play%.
    - **FC** (11%): High Contact% that tends to produce fouls more than balls in play.

Pitch Mix vs RHH (Results + Physics)

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
FF	43	43.9%	50.0%	20.9%	41.9%	5.6%	16.7%	94.4%	44.2%	32.6%
SL	30	30.6%	45.5%	36.7%	53.3%	50.0%	31.2%	50.0%	36.7%	30.0%
CH	13	13.3%	—	7.7%	61.5%	0.0%	62.5%	100.0%	61.5%	15.4%
CU	12	12.2%	25.0%	16.7%	33.3%	0.0%	50.0%	100.0%	58.3%	41.7%

Pitch Mix vs LHH (Results + Physics)

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
FF	44	44.0%	36.4%	15.9%	47.7%	4.8%	38.1%	95.2%	43.2%	34.1%
FC	22	22.0%	66.7%	4.5%	63.6%	7.1%	21.4%	92.9%	31.8%	27.3%
CH	17	17.0%	100.0%	29.4%	47.1%	25.0%	50.0%	75.0%	35.3%	29.4%
CU	15	15.0%	60.0%	33.3%	20.0%	33.3%	33.3%	66.7%	20.0%	40.0%
SL	2	2.0%	—	0.0%	0.0%	0.0%	0.0%	0.0%	50.0%	50.0%

- **Pitch Mix & Result vs RHH/LHH**
    - FF and SL are the main weapons against RHH, and SL has the best miss profile; FF is the only true primary weapon against LHH, while SL is barely used, and the other pitches are spread thin at similar, smaller shares.
    - FF has a 50% FPS%, and a low In-Play% while having 94.4% Contact%, an efficient strike-getting pitch against RHH, but the effectiveness is lower against LHH
    - CH and CU both show better CSW/Whiff and lower in-play vs LHH than vs RHH
    - FC is only pitched against LHH, and has a High Contact% that tends to produce fouls more than balls in play. A weapon to steal strikes against Lefties.

Two-Strike Finisher & First-Pitch Plan

Pitch Type: All Min sample (N ≥): 0 Update

Two-Strike Finisher — ALL

PitchType	Pitches	Usage%	Swing%	InPlay%	Zone%	Ball%	K%
FF	24	33.8%	62.5%	33.3%	45.8%	20.8%	33.3%
CH	18	25.4%	44.4%	62.5%	27.8%	38.9%	25.0%
SL	11	15.5%	27.3%	66.7%	27.3%	45.5%	60.0%
CU	10	14.1%	30.0%	33.3%	30.0%	40.0%	75.0%
FC	8	11.3%	37.5%	66.7%	25.0%	37.5%	50.0%

Note: For two-strike analysis, all process metrics are reported per pitch, while K% is reported per finisher.

- **Two-Strike Finisher**

- FF (34% use): Remains the primary pitch.
- SL (25%): Usage up from 16.2% → 25%; K% = 60% (per finisher).
- CH (16%), CU (14%), FC (11%): Usage remains similar to its overall situation.
- CH, SL, FC have high-risk In-Play%.

Two-Strike Finisher — vs RHH

PitchType	Pitches	Usage%	Swing%	InPlay%	Zone%	Ball%	K%
FF	14	37.8%	50.0%	14.3%	42.9%	28.6%	75.0%
CH	9	24.3%	55.6%	60.0%	44.4%	22.2%	40.0%
SL	9	24.3%	33.3%	66.7%	22.2%	44.4%	50.0%
CU	5	13.5%	40.0%	50.0%	60.0%	40.0%	50.0%

Two-Strike Finisher — vs LHH

PitchType	Pitches	Usage%	Swing%	InPlay%	Zone%	Ball%	K%
FF	10	29.4%	80.0%	50.0%	50.0%	10.0%	0.0%
CH	9	26.5%	33.3%	66.7%	11.1%	55.6%	0.0%
FC	8	23.5%	37.5%	66.7%	25.0%	37.5%	50.0%
CU	5	14.7%	20.0%	0.0%	0.0%	40.0%	100.0%
SL	2	5.9%	0.0%	0.0%	50.0%	50.0%	100.0%

- **Two-Strike Finisher vs RHH/LHH**
  - Splitting by side shows FF share drops, and vs LHH, the finishing mix is more evenly distributed across pitch types.
  - vs RHH: FF K% = 75% (per finisher) with In-Play% = 14.3% → dominant profile.
  - vs LHH: FF In-Play% = 50% (much higher risk than vs RHH).
  - Others: Either similar to baseline or sample too small to read.

First-Pitch Plan — ALL

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
FF	19	40.4%	42.1%	15.8%	31.6%	0.0%	16.7%	100.0%	36.8%	52.6%
SL	11	23.4%	45.5%	36.4%	36.4%	50.0%	25.0%	50.0%	36.4%	45.5%
CU	9	19.1%	44.4%	44.4%	11.1%	0.0%	100.0%	100.0%	55.6%	44.4%
FC	6	12.8%	66.7%	16.7%	66.7%	25.0%	0.0%	75.0%	16.7%	33.3%
CH	2	4.3%	100.0%	50.0%	100.0%	50.0%	0.0%	50.0%	50.0%	0.0%

- **First-Pitch Plan**
  - FF (40.4% use): Primary first pitch, but has the highest Ball% among all types.
  - SL (23.4%): Usage up from 16.2% → 23.4%, but Ball% 31.2% → 45.5%
  - CU (19.1%): Usage up from 13.6% → 19.1%, and remains high Ball%.
  - FC (12.8%): User remains similar to its overall situation, but has a 0 In-Play%.
  - CH (4.3%): N=2, not interpretable.

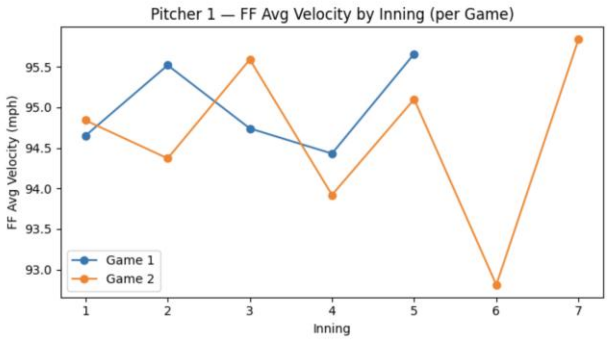
First-Pitch Plan — vs RHH

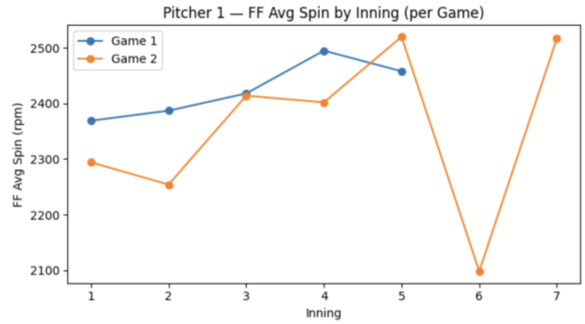
PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
SL	11	47.8%	45.5%	36.4%	36.4%	50.0%	25.0%	50.0%	36.4%	45.5%
FF	8	34.8%	50.0%	25.0%	25.0%	0.0%	0.0%	100.0%	50.0%	50.0%
CU	4	17.4%	25.0%	25.0%	25.0%	0.0%	100.0%	100.0%	50.0%	50.0%

First-Pitch Plan — vs LHH

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
FF	11	45.8%	36.4%	9.1%	36.4%	0.0%	25.0%	100.0%	27.3%	54.5%
FC	6	25.0%	66.7%	16.7%	66.7%	25.0%	0.0%	75.0%	16.7%	33.3%
CU	5	20.8%	60.0%	60.0%	0.0%	0.0%	0.0%	0.0%	60.0%	40.0%
CH	2	8.3%	100.0%	50.0%	100.0%	50.0%	0.0%	50.0%	50.0%	0.0%

- **First-Pitch Plan vs RHH/LHH**
  - vs RHH: SL is the top first-pitch choice (47.8% use) with Whiff% 50.0, Zone% 36.4, In-Play% 25.0, Ball% 45.5. FF shows FPS% 50.0 and In-Play% 0.0 (N=8).
  - vs LHH: FF is the main first-pitch option (45.8% use) with Whiff% 0.0, Ball% 54.5, In-Play% 25.0. FC and CU offer high FPS%, and In-Play% 0.0 (N=6,5).
  - CU across sides: Usage is similar; results are mixed with small samples (RHH In-Play% 25.0 vs LHH 0.0, N=4,5).



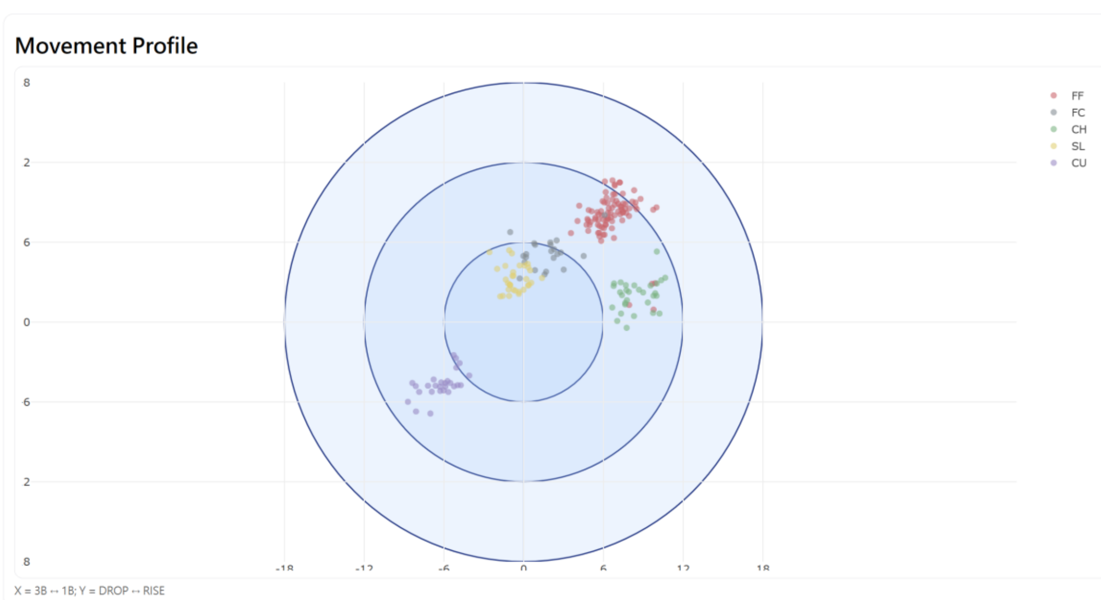


- **Stamina**
  - **Velo/Stuff held late:** Across both starts, the FF sat about 94~96 mph with no inning-to-inning drop, and spin stayed about 2,350~2,500 rpm. Pitcher 1 finished both games without a late-inning fade (small sample).

- **Pitcher 1 Summary**
  - **Results better than process:** 0.92 WHIP, .529 OPS; but FPS 48.9 / CSW 20.7 / Whiff 15.2 are light.
  - **Low FPS% + high Ball%** point to lose control/command. If the command tightens, pitches per batter faced should drop, helping him carry deeper innings.
  - **Stronger vs RHH**, and has more BB + fewer K when vs LHH, while having better FPS%, showing that the issue is finishing, not starting.
  - **Two-strike**: vs RHH the FF finishes well; vs LHH the FF In-Play ≈ 50% (higher risk); CH, SL, FC have high-risk In-Play%.
  - **First pitch**: Low In-Play% across all pitches except CU; FF is the most efficient strike-getter vs RHH, FC/CU are usable vs LHH. Pitcher 1 might consider pitching more FF as the first pitch against RHH, and mix more non-FF pitch types against LHH.

- **Recommendation**
  - Tighten command: FPS is low and Ball% is high across several first-pitch looks. Clean up the zone early (target 55~60% FPS) to cut Pitches/BF and extend outings.
  - With two strikes, keep FF as the finisher vs RHH, and shift some finishes to CH/FC vs LHH.
  - CU can be used more as the finisher and raise its usage with two strikes against RHH, and SL may be pitched more against LHH to test the effectiveness.
  - CH is a good strike-getting pitch, but should be used less, especially against RHH.

- **Notable Information: Movement Profile**



Other Starters: One Start (Pitcher 7&11)
1. Pitcher 7

Overview (Pitcher)

Box Score

Pitches	Balls	Strikes	H	HR	BB	K	IP	WHIP	OPS
75	26	30	4	0	2	5	6.0	1.00	0.544

Advanced Stats

Batter Faced	Pitches/BF	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	K%	BB%
26	2.88	42.3%	18.7%	48.0%	16.7%	52.8%	83.3%	19.2%	7.7%

Advanced Stats vs RHH

Batter Faced	Pitches/BF	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	K%	BB%
17	2.53	47.1%	20.9%	58.1%	16.0%	56.0%	84.0%	17.6%	0.0%

Advanced Stats vs LHH

Batter Faced	Pitches/BF	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	K%	BB%
9	3.56	33.3%	15.6%	34.4%	18.2%	45.5%	81.8%	22.2%	22.2%

Note: Per pitch: CSW%, Swing%; Per swing: Whiff%, In-Play%, Contact%; Per PA: K%, BB%, K–BB%

- **Game Stats**
  - **Elite:** 2.88 Pitches per Batter Faced, 1.00 WHIP, 0.544 OPS
  - **Quality:** 6 IP/G
  - **Average:** 7.7 BB%
  - **Below Average:** 19.2K%, 11.5 K-BB%, 42.3 FPS%, 18.7 CSW%, 16.7 Whiff%, 52.8 In-Play%, 83.3% Contact%

- **R/L Split (directionally meaningful; 1-game sample + RHH sample size is double LHH sample size):**
  - Pitches per Batter Faced: 2.53 vs RHH → 3.56 vs LHH (+1.03).
  - FPS%: 47.1% vs RHH → 33.3% vs LHH (-13.8 pts).
  - CSW% (per pitch): 20.9% → 15.6% vs LHH (−5.3 pts).
  - K% (per PA): 17.6% → 22.2% vs LHH (+8.7 pts).
  - BB% (per PA): 0.0% → 22.2% vs LHH (+22.2 pts).
  - Swing% (per pitch): 58.1% → 34.4% vs LHH (-23.7 pts).
  - In-Play% (per swing): 56.0% → 45.5% vs LHH (-10.5 pts).

- **In short:**
  - **Results > process:** 2.88 Pitches/BF, 1.00 WHIP, .544 OPS → good run prevention and efficiency, but FPS 42.3 / CSW 18.7 / Whiff 16.7 are light, and In-Play 52.8 / Contact 83.3 suggest the potential traffic risk.
  - **vs RHH (steadier):** FPS 47.1, Swing 58.1, K 17.6 / BB 0.0 → tempo under control with minimal walks.
  - **vs LHH (more volatile):** FPS 33.3 with a clear drop in Swing%; pitcher 7 also has K 22.2% / BB 22.2%; an 84.4% Contact% with only 45.5% In-Play% is good → half contacts are fouls, which keeps balls out of play but extends counts, a classic "miss bat or miss zone" profile with a high walk cost.

Pitch Mix & Results

Pitch Type: All Min sample (Pitches ≥): 0 Update

First row is TOTAL (pitcher-level). Below are per pitch type.

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
TOTAL	75	—	42.3%	18.7%	48.0%	16.7%	52.8%	83.3%	42.7%	30.7%
FF	21	28.0%	37.5%	14.3%	42.9%	0.0%	66.7%	100.0%	42.9%	38.1%
SI	20	26.7%	60.0%	15.0%	50.0%	10.0%	30.0%	90.0%	55.0%	20.0%
CU	13	17.3%	25.0%	23.1%	46.2%	33.3%	66.7%	66.7%	46.2%	38.5%
SL	11	14.7%	50.0%	27.3%	81.8%	33.3%	55.6%	66.7%	36.4%	9.1%
FC	10	13.3%	40.0%	20.0%	20.0%	0.0%	50.0%	100.0%	20.0%	50.0%

- **Pitch Mix & Result**
  - **FF** (28% use): A poor FPS%, CSW%, and a poor 0.0 Whiff%. In-play% is high.
  - **SI** (27%): A quality 60% FPS%. CSW% is low and has a 0% Whiff%, but In-Play% is also low.
  - **CU** (17%): A poor 25% FPS% but a Whiff 0.0%. In-play% is high.
  - **SL** (15%): The highest Whiff% & In-Play% and the lowest Ball%.
  - **FC** (13%): A poor 0% Whiff% while having 100% Contact%, and a 50% Ball%

Pitch Mix vs RHH (Results + Physics)

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
SI	14	32.6%	60.0%	21.4%	64.3%	11.1%	33.3%	88.9%	71.4%	7.1%
SL	10	23.3%	66.7%	30.0%	80.0%	37.5%	50.0%	62.5%	40.0%	10.0%
FC	8	18.6%	40.0%	25.0%	12.5%	0.0%	100.0%	100.0%	25.0%	62.5%
FF	7	16.3%	0.0%	0.0%	57.1%	0.0%	75.0%	100.0%	57.1%	42.9%
CU	4	9.3%	50.0%	25.0%	75.0%	0.0%	100.0%	100.0%	75.0%	0.0%

Pitch Mix vs LHH (Results + Physics)

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
FF	14	43.8%	50.0%	21.4%	35.7%	0.0%	60.0%	100.0%	35.7%	35.7%
CU	9	28.1%	0.0%	22.2%	33.3%	66.7%	33.3%	33.3%	33.3%	55.6%
SI	6	18.8%	—	0.0%	16.7%	0.0%	0.0%	100.0%	16.7%	50.0%
FC	2	6.3%	—	0.0%	50.0%	0.0%	0.0%	100.0%	0.0%	0.0%
SL	1	3.1%	0.0%	0.0%	100.0%	0.0%	100.0%	100.0%	0.0%	0.0%

- **Pitch Mix & Result vs RHH/LHH**
  - SI and SL are the main weapons against RHH; both have great FPS%, but SL has a 50% In-play% while having a 37.5% Whiff%.
  - FF is the primary weapon against LHH, with a 50% FPS%, but also has a 100% Contact% per swing; CU is the secondary weapon against LHH, with a high Whiff% but also a high Ball%.

Two-Strike Finisher & First-Pitch Plan

Pitch Type: All Min sample (N ≥): 0 Update

Two-Strike Finisher — ALL

PitchType	Pitches	Usage%	Swing%	InPlay%	Zone%	Ball%	K%
SI	4	33.3%	25.0%	0.0%	75.0%	25.0%	100.0%
CU	3	25.0%	33.3%	100.0%	0.0%	33.3%	50.0%
SL	3	25.0%	33.3%	100.0%	33.3%	33.3%	50.0%
FF	2	16.7%	50.0%	100.0%	100.0%	0.0%	50.0%

Note: For two-strike analysis, all process metrics are reported per pitch, while K% is reported per finisher.

- **Two-Strike Finisher (Limited Sample)**
  - Distributed across SI/SL/CU/FF (2 to 4 pitches each) → No clear primary in this game.
  - SI shows 0 In-Play% and 100% K% in this sample.
  - Other pitches have a 100% In-Play%.

Two-Strike Finisher — vs RHH

PitchType	Pitches	Usage%	Swing%	InPlay%	Zone%	Ball%	K%
SL	3	42.9%	33.3%	100.0%	33.3%	33.3%	50.0%
SI	2	28.6%	0.0%	0.0%	100.0%	0.0%	100.0%
CU	1	14.3%	100.0%	100.0%	0.0%	0.0%	0.0%
FF	1	14.3%	100.0%	100.0%	100.0%	0.0%	0.0%

Two-Strike Finisher — vs LHH

PitchType	Pitches	Usage%	Swing%	InPlay%	Zone%	Ball%	K%
CU	2	40.0%	0.0%	0.0%	0.0%	50.0%	100.0%
SI	2	40.0%	50.0%	0.0%	50.0%	50.0%	—
FF	1	20.0%	0.0%	0.0%	100.0%	0.0%	100.0%

- **Two-Strike Finisher vs RHH/LHH (Limited Sample)**
  - vs RHH: Pitcher 7 split his two-strike pitches fairly evenly among SL, SI, CU, and FF.
  - vs LHH: Only distributed pitches among CU, SI, and FF
  - Sample is too small, but has a high In-Play% against RHH while 0 In-Play% against LHH.

First-Pitch Plan — ALL

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
FF	8	30.8%	37.5%	25.0%	25.0%	0.0%	50.0%	100.0%	37.5%	50.0%
FC	5	19.2%	40.0%	40.0%	0.0%	0.0%	0.0%	0.0%	20.0%	60.0%
SI	5	19.2%	60.0%	40.0%	60.0%	0.0%	66.7%	100.0%	60.0%	0.0%
CU	4	15.4%	25.0%	25.0%	25.0%	0.0%	100.0%	100.0%	50.0%	50.0%
SL	4	15.4%	50.0%	25.0%	100.0%	25.0%	50.0%	75.0%	25.0%	0.0%

- **First-Pitch Plan (Limited Sample)**
  - FF (31% use): Primary first pitch, but only has 37.5% FPS% with a 50% Ball%
  - All other pitch types have even splits. SI has the highest 60% FPS%; FC and CU have high Ball%
  - Overall: Whiff% is low, and only FC has a lower In-Play%.

First-Pitch Plan — vs RHH

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
FC	5	29.4%	40.0%	40.0%	0.0%	0.0%	0.0%	0.0%	20.0%	60.0%
SI	5	29.4%	60.0%	40.0%	60.0%	0.0%	66.7%	100.0%	60.0%	0.0%
SL	3	17.6%	66.7%	33.3%	100.0%	33.3%	33.3%	66.7%	33.3%	0.0%
CU	2	11.8%	50.0%	50.0%	50.0%	0.0%	100.0%	100.0%	100.0%	0.0%
FF	2	11.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%

First-Pitch Plan — vs LHH

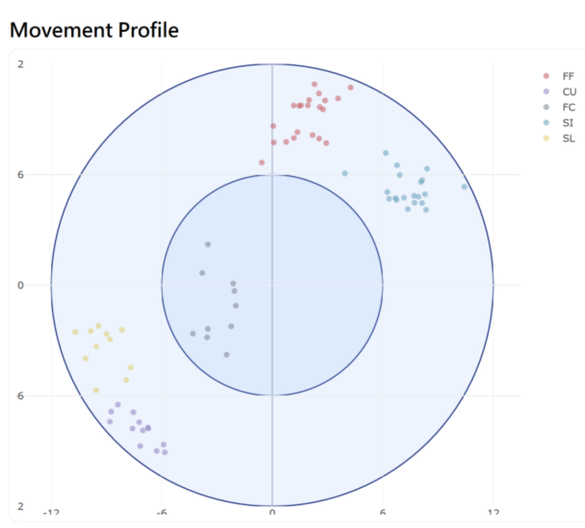
PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
FF	6	66.7%	50.0%	33.3%	33.3%	0.0%	50.0%	100.0%	50.0%	33.3%
CU	2	22.2%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%
SL	1	11.1%	0.0%	0.0%	100.0%	0.0%	100.0%	100.0%	0.0%	0.0%

- **First-Pitch Plan vs RHH/LHH (Limited Sample)**
  - vs RHH: FC and SI are the primary choices, while others have even distribution. FC has a high 60% Ball%, while SI and SL have 60+ FPS% and 0 Ball%
  - vs LHH: FF is the main first-pitch option (67% use) with a slightly better FPS%.

- **Pitcher 7 Summary**
  - **Results and process:** 2.88 Pitches/BF, 1.00 WHIP, .544 OPS; but FPS 42.3 / CSW 18.7 / Whiff 16.7 are light with In-Play 52.8 / Contact 83.3.
  - **vs RHH steadier**: FPS 47.1, K 17.6 / BB 0.0, higher Swing%.
  - **vs LHH more volatile**: FPS 33.3, Swing% down, K 22.2 / BB 22.2.
  - **Pitch mix (all)**: FF shows low FPS/CSW and high In-Play; SI 27% has FPS 60% with lower In-Play but 0 Whiff; SL 15% carries the best Whiff with the lowest Ball%; CU/FC show higher Ball%.
  - **First pitch (limited sample)**: FF is primary, but FPS 37.5 / Ball 50.0. SI posts the best FPS (60.0).
    - vs RHH: FC & SI lead; SI/SL both 60%+ FPS with 0 Ball%.
    - vs LHH: FF is main (67% use) with ~50% FPS.
  - Two-strike (limited): Finisher split across SI/SL/CU/FF (2 to 4 pitches each); SI shows 0 In-Play and 100% K% in this sample.

- **Recommendation**
  - Tighten early count command: FPS is low, and Ball% runs high in several first-pitch looks.
  - High In-Play% shows high risks, may not consider a great performance with low WHIP and OPS.
  - First pitch
    - vs RHH: Lean on SI/SL for get-ahead (60%+ FPS, 0 Ball%); use FC selectively (Ball 60%); keep FF sparing in this look (tiny N, FPS 0%).
    - vs LHH: Keep FF primary (~50% FPS); sprinkle CU carefully (miss potential with higher Ball%); other types are very small-N.
  - Two-strike
    - No clear primary in this game. SL is the best miss pitch (best Whiff) and monitor K% (per finisher) and In-Play% (per pitch) by side.
  - **Sample caution:** With only one game and no batted-ball quality, it's hard to tell whether the low In-Play% is true "stuff" or just balls hit at defenders. → should not draw firm conclusions.

- **Notable Information: Movement Profile**



2. Pitcher 11

Overview (Pitcher)

Box Score

Pitches	Balls	Strikes	H	HR	BB	K	IP	WHIP	OPS
97	33	44	8	0	1	5	6.0	1.50	0.706

Advanced Stats

Batter Faced	Pitches/BF	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	K%	BB%
26	3.73	38.5%	18.6%	47.4%	10.9%	43.5%	89.1%	19.2%	3.8%

Advanced Stats vs RHH

Batter Faced	Pitches/BF	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	K%	BB%
17	4	47.1%	20.6%	45.6%	12.9%	35.5%	87.1%	29.4%	5.9%

Advanced Stats vs LHH

Batter Faced	Pitches/BF	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	K%	BB%
9	3.22	22.2%	13.8%	51.7%	6.7%	60.0%	93.3%	0.0%	0.0%

Note: Per pitch: CSW%, Swing%; Per swing: Whiff%, In-Play%, Contact%; Per PA: K%, BB%, K–BB%

- **Game Stats**
  - **Elite:** 3.8 BB%
  - **Quality:** 6 IP/G, 3.73 Pitches per Batter Faced
  - **Average:** 15.4 K%-BB%, .706 OPS, 43.5 In-Play%
  - **Below Average:** 19.2 K%, 1.50 WHIP, 38.5 FPS%, 18.6 CSW%, 10.9 Whiff%, 89.1 Contact%

- **R/L Split (directionally meaningful; 1-game sample + RHH sample size is double LHH sample size):**
  - FPS%: 47.1% vs RHH → 22.2% vs LHH (-24.9 pts).
  - CSW% (per pitch): 20.6% → 13.8% vs LHH (−6.4 pts).
  - Whiff% (per swing): 12.9% → 6.7% vs LHH (-6.2 pts).
  - In-Play% (per swing): 35.5% → 60.0% vs LHH (+25.5 pts).
  - K% (per PA): 29.4% → 0% vs LHH (-29.4 pts).
  - BB% (per PA): 5.9% → 0% vs LHH (-5.9 pts).
  -
- **In short:**
  - **Process is light** (low FPS/CSW/Whiff, high Contact). 8 hits inflate WHIP 1.50, but OPS .706 suggests the damage was singles-heavy.
  - **vs RHH (better):** K%–BB% 23.5 with low In-Play% (35,5) → solid run prevention vs righties.
  - **vs LHH (weaker):** Whiff 6.7% with In-Play% 60.0 → lack of effectiveness in play vs lefties.

Pitch Mix & Results

Pitch Type: All Min sample (Pitches ≥): 0 Update

First row is TOTAL (pitcher-level). Below are per pitch type.

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
TOTAL	97	—	38.5%	18.6%	47.4%	10.9%	43.5%	89.1%	39.2%	32.0%
CU	30	30.9%	16.7%	10.0%	46.7%	14.3%	42.9%	85.7%	26.7%	36.7%
FF	23	23.7%	71.4%	30.4%	65.2%	20.0%	33.3%	80.0%	60.9%	13.0%
SI	22	22.7%	14.3%	4.5%	59.1%	0.0%	53.8%	100.0%	45.5%	31.8%
SL	11	11.3%	66.7%	27.3%	18.2%	0.0%	50.0%	100.0%	27.3%	45.5%
CH	11	11.3%	33.3%	36.4%	18.2%	0.0%	50.0%	100.0%	27.3%	45.5%

- **Pitch Mix & Result**
  - **CU** (31% use): Low FPS%, CSW%, Whiff%. An average In-Play%.
  - **FF** (24%): A splendid 71.4 FPS% with only 13.0 Ball% → Best strike getter
  - **SI** (23%): Lowest 4.5 CSW%.
  - **SL** (11%) & **CH** (11%): Both have 45.5 Ball% and 0 Whiff%. SL has a great 66.7 FPS%, but CH only has 33.3 FPS%
  - **Overall:** Ball% is high for all pitches except FF; In-Play% is around average performance.

Pitch Mix vs RHH (Results + Physics)

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
FF	23	33.8%	71.4%	30.4%	65.2%	20.0%	33.3%	80.0%	60.9%	13.0%
CU	21	30.9%	0.0%	4.8%	42.9%	11.1%	22.2%	88.9%	23.8%	38.1%
CH	11	16.2%	33.3%	36.4%	18.2%	0.0%	50.0%	100.0%	27.3%	45.5%
SL	8	11.8%	66.7%	25.0%	25.0%	0.0%	50.0%	100.0%	25.0%	37.5%
SI	5	7.4%	0.0%	0.0%	60.0%	0.0%	66.7%	100.0%	40.0%	20.0%

Pitch Mix vs LHH (Results + Physics)

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
SI	17	58.6%	16.7%	5.9%	58.8%	0.0%	50.0%	100.0%	47.1%	35.3%
CU	9	31.0%	33.3%	22.2%	55.6%	20.0%	80.0%	80.0%	33.3%	33.3%
SL	3	10.3%	—	33.3%	0.0%	0.0%	0.0%	0.0%	33.3%	66.7%

- **Pitch Mix & Result vs RHH/LHH**
  - **vs RHH:** FF and CU are the main weapons.
    - FF: 71.4 FPS%, 30.4 CSW%, average Whiff%, lowest Ball% among all pitches.
    - CU: high Ball%, low Whiff%, and In-Play% 22.2 shows great traffic prevention.
    - CH is the 3rd-most vs RHH, Ball% high, Whiff% low.
    - SL shows 66.7 FPS% (small N)
  - **vs LHH**
    - Only SI/CU/SL were used. SI is primary (58.6% use) with very low FPS% and high Ball%, though In-Play% is lower than CU, it's still high → Limited control/impact vs LHH.

Two-Strike Finisher & First-Pitch Plan

Pitch Type: All Min sample (N ≥): 0 Update

Two-Strike Finisher — ALL

PitchType	Pitches	Usage%	Swing%	InPlay%	Zone%	Ball%	K%
CU	13	59.1%	46.2%	50.0%	30.8%	23.1%	50.0%
FF	4	18.2%	75.0%	66.7%	75.0%	0.0%	33.3%
SL	3	13.6%	33.3%	0.0%	0.0%	33.3%	100.0%
SI	2	9.1%	100.0%	100.0%	50.0%	0.0%	0.0%

Note: For two-strike analysis, all process metrics are reported per pitch, while K% is reported per finisher.

- **Two-Strike Finisher**
  - CU is the primary finisher, as other pitch types appear sparsely and are evenly distributed.
  - CU: K% 50 (per finisher) with In-Play 50.0(per pitch); others are too small to interpret.

Two-Strike Finisher — vs RHH

PitchType	Pitches	Usage%	Swing%	InPlay%	Zone%	Ball%	K%
CU	11	55.0%	36.4%	25.0%	27.3%	27.3%	75.0%
FF	4	20.0%	75.0%	66.7%	75.0%	0.0%	33.3%
SL	3	15.0%	33.3%	0.0%	0.0%	33.3%	100.0%
SI	2	10.0%	100.0%	100.0%	50.0%	0.0%	0.0%

Two-Strike Finisher — vs LHH

PitchType	Pitches	Usage%	Swing%	InPlay%	Zone%	Ball%	K%
CU	2	100.0%	100.0%	100.0%	50.0%	0.0%	0.0%

- **Two-Strike Finisher vs RHH/LHH (Limited Sample)**
  - This game provided very few two-strike looks vs LHH (only 2 pitches); both resulted in balls in play and no strikeouts. Not enough to judge two-strike effectiveness vs lefties

First-Pitch Plan — ALL

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
FF	7	26.9%	71.4%	42.9%	57.1%	25.0%	25.0%	75.0%	57.1%	14.3%
SI	7	26.9%	14.3%	0.0%	14.3%	0.0%	0.0%	100.0%	14.3%	85.7%
CU	6	23.1%	16.7%	16.7%	16.7%	0.0%	100.0%	100.0%	33.3%	66.7%
CH	3	11.5%	33.3%	33.3%	0.0%	0.0%	0.0%	0.0%	33.3%	66.7%
SL	3	11.5%	66.7%	66.7%	0.0%	0.0%	0.0%	0.0%	33.3%	33.3%

- **First-Pitch Plan**
  - FF / SI / CU are the primary first-pitch options (~27% use each).
  - FF has the best FPS% = 71.4. Others: SI 14.3 (lowest), CU 23.1 (low), CH 33.3 (mid), SL 66.7

(small N).
  - Overall: Ball% is high for most pitches except FF, but most pitches have low In-Play% except CU.

First-Pitch Plan — vs RHH

PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
FF	7	41.2%	71.4%	42.9%	57.1%	25.0%	25.0%	75.0%	57.1%	14.3%
CH	3	17.6%	33.3%	33.3%	0.0%	0.0%	0.0%	0.0%	33.3%	66.7%
CU	3	17.6%	0.0%	0.0%	33.3%	0.0%	100.0%	100.0%	33.3%	66.7%
SL	3	17.6%	66.7%	66.7%	0.0%	0.0%	0.0%	0.0%	33.3%	33.3%
SI	1	5.9%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%

First-Pitch Plan — vs LHH

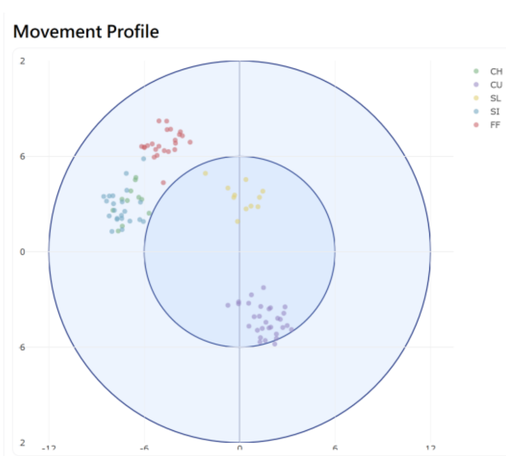
PitchType	Pitches	Usage%	FPS%	CSW%	Swing%	Whiff%	InPlay%	Contact%	Zone%	Ball%
SI	6	66.7%	16.7%	0.0%	16.7%	0.0%	0.0%	100.0%	16.7%	83.3%
CU	3	33.3%	33.3%	33.3%	0.0%	0.0%	0.0%	0.0%	33.3%	66.7%

- **First-Pitch Plan vs RHH/LHH**
  - **vs RHH**: FF is the primary choice; CH/CU/SL are secondary with similar shares, SI minimal. FF & SL (small N) show higher FPS; CH/CU/SI carry higher Ball%.
  - **vs LHH (small sample):** SI primary, CU secondary; both show very high Ball%.

- **Pitcher 11 Summary**
  - **Results vs process:** 3.73 Pitches/BF (quality), 3.8 BB% (elite); but WHIP 1.50 (below avg) with FPS 38.5 / CSW 18.6 / Whiff 10.9 light and Contact 89.1 high. OPS .706 suggests the 8 hits were mostly singles.
  - **vs RHH steadier:** FPS 47.1, In-Play 35.5, K-BB% 23.5.
  - **vs LHH weaker:** FPS 22.2, Whiff 6.7, In-Play 60.0, K 0 / BB 0 (balls in play, no punchouts).
  - **Pitch mix:** FF (24%) is the best strike-getter (FPS 71.4, Ball 13.0). SI (23%) carries the weakest process (FPS 14.3, CSW 4.5). CU/CH/SL show high Ball%; SL small-N but FPS 66.7.
  - **Two-strike (limited):** CU is the primary finisher with a 50% K% 50 but 50.0% In-Play%; Very few LHH two-strike looks (2 pitches); Overall, two-strike pitches have high In-Play%.
  - **First pitch:** FF/SI/CU are the primary options (~27% each). FF is the best get-ahead pitch (FPS 71.4, Ball 13.0); SI 14.3 FPS and CU 23.1 FPS carry higher Ball%. vs RHH FF leads; vs LHH SI leads with very high Ball%.

- **Recommendation**
  - **Tighten early count command:** Target 55–60% FPS; overall Ball% is high outside of FF.
  - **First pitch**
    - Keep FF primary when vs RHH; use CH/SL sparingly given higher Ball%.
    - Prevent from using CU, as it has a 100% In-Play%.
  - **Two-strike**
    - Keep CU as the default finisher vs RHH, but watch In-Play; log by side and adjust as more LHH data arrives.
  - May consider adding FF into the mix when vs LHH, as the current mix (SI/CU/SL) shows poor strike entry → Need further evaluation as there is no FF vs LHH data in this game.
  - **Sample caution:** With only one game and no batted-ball quality, it's hard to tell whether the low In-Play% is true "stuff" or just balls hit at defenders. → should not draw firm conclusions.

- **Notable Information: Movement Profile**



Batter Evaluation

Scope

- Since we do not have batted-ball data (no EV/LA), I only looked at hitters with PA ≥ 5.
- All stats are outcome-based: OBP, SLG, Pitches/PA, two-strike, and early-PA.

Tag rules

- on-base hot: OBP ≥ .400
- impact: SLG ≥ .500
- grindy: Pitches/PA ≥ 4.2 (works the count)
- aggressive early: Early-PA% ≥ 50% (PA ends in first 3 pitches)
- 2S survive: Two-Strike OBP ≥ .300 (still reaches after two strikes)

BatterId	PA	Pitches/PA	OBP	SLG	K%	BB%	H%	Two-Strike Reached	Two-Strike Hit%	Two-Strike OB	Early-PA %	Early-PA OB	Tags
2	5	4.2	0	0	80	0	0	20	0	0	40	0	grindy
4	9	3.67	0.22	0.33	0	0	22	22.2	0	0	55.6	0.2	aggressive early
7	5	4	0.4	0.5	20	0	40	40	50	0.5	40	0.5	on-base hot, impact, 2S survive
13	9	4.22	0.22	0.8	33	11.1	11	11.1	0	0	11.1	1	impact, grindy
14	5	3.4	0.4	1	20	0	40	40	50	0.5	60	0.333	on-base hot, impact, aggressive early, 2S survive
17	5	5	0.4	0.33	20	20	20	60	0	0	20	1	on-base hot, grindy
18	7	4.14	0.29	0.25	29	14.3	14	14.3	0	0	42.9	0	
19	10	3.3	0.2	0.33	40	0	20	20	0	0	60	0.333	aggressive early
20	5	3.6	0.2	0.8	0	0	20	40	50	0.5	60	0	impact, aggressive early, 2S survive
21	10	3.6	0.6	0.63	10	10	50	10	0	0	50	0.8	on-base hot, impact, aggressive early
24	10	3.1	0.4	0.4	0	0	40	20	50	0.5	70	0.429	on-base hot, aggressive early, 2S survive
26	5	2.2	0	0	20	0	0	0	0	0	80	0	aggressive early
29	5	3.8	0	0	40	0	0	40	0	0	20	0	
33	5	4.6	0	0	40	0	0	20	0	0	40	0	grindy
34	9	3.22	0.56	0.83	11	22.2	33	11.1	0	0	55.6	0.4	on-base hot, impact, aggressive early

Quick takeaways (from the PA ≥ 5 table)

- A few hitters show both on-base and impact (e.g., #21, #34, #14, #7). Good run creators in this sample.
- Many are aggressive early (#4, #14, #20, #21, #24, #26, #29, #34). Some turn that into results (#21, #24, #34). Others are early but lighter in OBP/SLG.
- Grindy hitters who push pitch counts: #2, #13, #18, #33.
- Two-strike skill shows for #7, #14, #24 (tagged 2S survive). A few others drop off with two strikes.